Qin 2023 vs Armitage 2021 (IV-MBM EQP v2.0) on PFBA, PFHxA, PFHpA, PFOA, PFNA

This version supersedes the earlier httk-based analysis in this file's history. Armitage's Supplementary Section S1, the Fragki main paper and the Rosenmai supporting information are now in hand, and two of the earlier conclusions were artefacts of substituting EPA's `httk` for the published parameterisation. What changed, and why, is in §2.

Second correction, 2026-10-20 (§15). Fragki's exposure medium is 2% FBS, not the 10% of its growth medium. Every number below is on the corrected value; the pre-correction figures are in this file's history.

Reproduce: `python3 tk_review/scripts/mbm_published_equations.py`, `python3 tk_review/scripts/mbm_medium_composition.py` and `python3 tk_review/scripts/make_mbm_figure.py`. Outputs: `db/mbm_published_equations.csv`, `figures/fig10_mbm_qin_vs_armitage.png`.

1. The two equations, as published

Qin (confirmed by his own text):

C_free = C_nom x V_tot / (V_w + D_BSA/w x V_protein,medium + D_lip/w x V_lipid,medium + D_SP/w x V_protein,cell + D_lip/w x V_lipid,cell)

`D_SP/w = D_non-specific`: *"D_cell/w was predicted by the MBM (Eq. 17) with D_non-specific (Table 1) as a surrogate for partitioning to structural proteins and D_lip/w (Table 2) for lipid."* There is **no cell-water term**: *"we considered only proteins and lipids of the cell homogenate as the main sorptive phase for D_cell/w, which is different from Henneberger et al. where the cells were recognized as an entity containing water, protein and lipid and all of them contributed."*

Armitage Eq. 2, with the SI S1 spLFERs:

C_W = M_T / (V_W + D_AW V_A + D_SaW V_Sa + D_SlW V_Sl + D_DsW V_Ds + D_CW V_C + D_PlW A_Pl)

log K_MW = 1.01 log Kow,N + 0.12 log K_SaW = 1.08 log Kow,N - 0.70 (log Kow,N = 4.5) d_OW = d_PlW = 3.1, d_MW = 1.0, d_SaW = 0.0 for acids (Table S1)

2. What the real SI changed, against the httk stand-in

	httk `armitage.R`	**Armitage 2021 SI**	**effect**
cell composition	one generic cell, membrane lipid **0.04**	**cell-line specific**; the HepG2-derived AREc2 line is 0.00046 / **0.00414** / 0.0514 / 0.944 (Table S4)	**9.1×**
albumin spLFER	`0.71 logKow + 0.42`	piecewise `1.08 logKow - 0.70` / `0.37 logKow + 2.56`	1.6×
ionisation offsets	d_OW = d_PlW = 3.5	**3.1**	small
anionic phospholipids, lysosomes	present	absent from the published model	small

The cell-composition term dominates, because Armitage's PFAS cell is almost entirely membrane lipid and the hepatic line has ten times less of it than the generic default. On Rosenmai PFOA at 100 µM: published 0.660% of dose in cells, with httk's generic cell 6.007%, measured 0.330%.

Two independent checks that the composition table is the right one to use: Armitage's PPAR*gamma* line (HEK293H — Qin's own cell line) is listed at **0.08** protein by volume against Qin's measured **0.0814**; and Armitage's AREc2 at 10 ng/cell × 5.14% protein × 1.35 g/mL gives **694 pg protein per cell** against the project spreadsheet's **675 pg/cell**.

3. Why the two models behave differently

They put the PFAS in different places, and nearly agree by coincidence. At a common C_w = 1 µM:

	nCF	Qin D_cell/w	Armitage D_cell/w	ratio	Qin D_BSA/w	Armitage D_SaW	ratio
PFBA	3	2.4	1.2	1.99	520	138	3.76
PFHxA	5	25.4	10.1	2.52	4 952	6 503	0.76
PFHpA	6	83.2	56.4	1.47	12 044	26 501	0.45
PFOA	7	272	337	0.81	24 298	51 615	0.47
PFNA	8	892	2 036	0.44	45 849	100 531	0.46

The two cell ratios land within 0.44–2.5× of each other — but for unrelated reasons:

- **Qin's cell is 92–95% protein** (the measured D_non-specific) and 5–8% lipid, with no water at all.
- **Armitage's cell is 79% water at PFBA**, flipping to **99.7% membrane lipid** by PFNA. Its bulk-protein term contributes 0.1–0.2%, because protein is treated as 3.5% of octanol and the anion then takes the -3.1 penalty.

So the one phase Qin says carries nearly all the PFAS is the one phase Armitage's cell effectively ignores. The numerical agreement across these five compounds is a coincidence of two wrong-looking routes crossing.

The real divergence is now in albumin, not the cell. Armitage's piecewise spLFER rises 1.08 per log Kow below 4.5 — 0.83 per CF2 — then flattens to 0.285 per CF2 above it. Qin's measured affinity rises smoothly at ~0.49 per CF2. The kink falls between PFBA and PFHxA, so Armitage reads 3.8× *low* at PFBA and 2.1–2.2× *high* from PFHpA up. Since albumin holds 19–98% of the dose in these assays, that is what moves the predictions.

4. Dose dependence: the one test Armitage cannot pass

Fitting C_cell ∝ C_nom^b:

series	measured	Qin	Armitage
Rosenmai PFBA	0.85	1.15	1.00
Rosenmai PFHxA	0.87	1.58	1.00
Rosenmai PFOA	1.14	1.82	1.00
Rosenmai PFNA	1.31	1.81	1.00
Fragki PFOA	1.42	1.49	1.00
Fragki PFNA	1.42	1.62	1.00

Armitage is linear by construction and is pinned at exactly 1.00. The measured exponent rises monotonically with chain length, 0.85 -> 1.42, and Qin reproduces that ordering, overshooting by 0.30–0.71 on Rosenmai and by only 0.07–0.20 on Fragki. This is the only place where Qin's extra machinery buys something Armitage structurally cannot deliver.

It is also the one thing the §15 medium correction *improves*: on 2% FBS Qin's Fragki exponents fall from 1.71 and 1.66 to 1.49 and 1.62 against a measured 1.42. Less albumin means less of Qin's Langmuir saturation, and the curvature it was adding was partly an artefact of a sink five times too big.

The two models' error still swaps sign with dose, though the correction has moved Qin's low-dose advantage onto Armitage:

	<=25 µM	>25 µM
Armitage	6.08×	**2.94×**
Qin	5.99×	8.63×

5. Performance

Fraction of dose in cells, n = 29:

	Armitage / measured	Qin / measured		
PFBA	1.2–1.7×	1.8–3.4×		
PFHxA	0.8–3.3×	2.5–40.5×		
PFHpA	1.0–4.3×	2.8–50.5×		
PFOA	2.0–26.3×	1.7–32.9×		
PFNA	4.2–14.6×	2.4–10.6×		
overall bias	**3.87×**	7.52×		
**overall \	error\	**	**4.02×**	7.52×

Implemented from its own SI, Armitage is the better predictor overall and is remarkably good on Rosenmai's short chains (0.8–1.7×). Both models now over-predict everywhere; what used to look like a low-dose regime where Qin won was Fragki's dose series being run on five times too much albumin (§15).

The error is no longer spread evenly across compounds. PFBA–PFHpA, which only Rosenmai and Amstutz measure, are unchanged at 1.2–1.6× for Armitage. PFOA and PFNA, the two compounds Fragki contributes, carry almost the whole residual.

Qin's parameters are measured only for PFBA and PFOA; PFHxA and PFHpA are interpolated between those two carboxylate anchors and PFNA is extrapolated beyond both (open symbols in the figure). Its two worst compounds, PFHxA and PFHpA, are interpolated ones.

6. Why Rosenmai stands out: the axis label, not the measurement

Rosenmai's Abstract says **"4–2500 ng mg^{-1} protein"**; Results §3.3 and the Fig. 2A axis say **nmol mg^{-1} protein**. The full Results sentence settles that these are the same numbers:

> *"The cellular PFAS concentrations, expressed as nmol mg^{-1} protein, for the 14 PFASs at treatment concentrations of 10, 30 and 100 muM are shown in Figure 2(A). The concentrations ranged 18–2490 nmol mg^{-1} protein for the PFCAs, 4–494 nmol mg^{-1} protein for the PFSAs, and were 81 and 435 nmol mg^{-1} protein for FOSA."*

The union of those three sub-ranges is **4 to 2490** — minimum 4 from the PFSAs, maximum 2490 from the PFCAs. The Abstract's range is **4 to 2500**. Same minimum, same maximum to rounding. One set of numbers is being reported under two different units, so **exactly one label is a typo and no reading makes both correct**. (The Abstract is therefore not an independent vote for ng/mg — it is the same data. Which label is wrong has to be settled on physical grounds.)

Four checks say the numbers are ng/mg.

(a) Rosenmai against Amstutz. Same cell line, same dose, same stated unit:

at 100 µM	Rosenmai	Amstutz	ratio	Rosenmai ÷ MW	ratio
PFHxA	164	0.289	568×	0.522	1.81×
PFHpA	297	0.326	911×	0.816	2.50×
PFOA	525	1.815	289×	1.268	0.70×
PFNA	458	2.237	205×	0.987	0.44×

(b) Rosenmai against itself. Its uptake-% table and its Fig. 2A values are the same measurement normalised differently, so the medium volume reconciling them must be one number for all 15 rows. Read as ng/mg it is 1.03–2.20 mL for the three pooled wells; read as nmol/mg it is 423–645 mL. A Corning 12-well is 22.1 mm across and ~17.5 mm deep, so one well brim-full is 6.7 mL and the whole plate is 81 mL: the ng/mg reading needs 5–11% of three wells, the nmol/mg reading needs **5–8× the entire plate**. It is the absolute magnitude that discriminates, and it does so without anyone knowing the true working volume.

A check that does NOT work, recorded so it is not repeated. The implied volume is constant within each compound across doses (PFOA 1.39 / 1.41 / 1.44 mL over a 10× range). An earlier version of this section called that tightness decisive. It is not evidence at all: the two readings differ by a constant molecular weight per compound, so they scale

identically with dose and the dose consistency is the same under either. Nor can reading one value off the figure settle it — PFOA at 100 µM reads 525, which under ng/mg implies 0.47 mL/well and under nmol/mg implies 195 mL/well; without an independently known working volume that observation only re-derives the volume, it does not test the unit.

A residual that survives the fix. Between compounds the implied volume runs 2.14× (PFBA 2.11 mL down to PFNA 1.15 mL), decreasing monotonically with chain length. Rosenmai's Table S2 internal-standard recoveries of 78–98% span only 1.26×, so something else is also going on in that paper.

(c) Scale, by sub-class. Back-calculating every compound in Rosenmai's Table 1 at the PFOA-implied volume gives PFCAs 12–1510 ng/mg and PFSAs 8–96 ng/mg, against the stated 18–2490 and 4–494. The same scale. Read as nmol/mg the stated minima are 330× (PFCA) and 520× (PFSA) above what the uptake table supports. The PFSA minimum of 4 is the sharpest of these, because PFOS has the lowest uptake of any compound in Table 1 (0.04–0.05%) and so pins the bottom of the range.

The FOSA entry corroborates that these are the Fig. 2A numbers: it has only *two* values, 81 and 435, because 100 µM was dropped for FOSA on cytotoxicity grounds — so Fig. 2A carries two FOSA bars, not three.

(d) Physical ceiling. As nmol/mg, Fig. 2A puts PFOA at 253 mM and PFNA at 266 mM inside the cell — a quarter of the cell by mass.

(e) A second label error in the same paper. The supporting information's Figure S2 gives total protein per well in **ng/cm^2**, with bars at a few hundred. A 12-well is 3.5 cm^2, so that reads as a few micrograms of protein per well, about 40× too little for 1.8 × 10^5 HepG2 cells; as µg/cm^2 it is ~20× too much. Whichever way it is read it cannot be right as printed, which establishes that unit labels in this paper are not reliable.

Reading the axis as nmol/mg multiplies C_cell by exactly the molecular weight, **214× (PFBA) to 464× (PFNA)** — 2.33–2.67 decades — and because the factor tracks MW it also tilts the chain-length trend by 0.34 decades.

On a fraction-of-dose basis Rosenmai does not stand out at all. The measurement is sound; the conversion to concentration is where the outlier is created.

7. Which axis to plot

Every source agrees on **protein per cell** — Armitage's AREc2 gives 694 pg/cell, the project spreadsheet 675 pg/cell — and disagrees **3.3× on cell volume** (10 pL vs 3.04 pL). Since all three datasets report per mg protein, the *fraction of dose in cells* is insensitive to that disagreement while *C_cell* carries all of it. Model discrimination should be done on the fraction.

8. Geometry, now sourced rather than assumed

	plate	medium	cells	FBS
Rosenmai	12-well × 3 pooled	3 × 1 mL (assumed; §6b implies ~0.7 mL/well)	5.43 × 10^5	10%
Amstutz	96-well	250 µL	4 × 10^4	10% (verified)
Fragki	24-well	**550 µL** (stated)	seeded 55 000/well, differentiated	**2%** (verified, §15)

Fragki's main paper confirms the plate and volume I had assumed. The FBS it does not confirm: exposure was in "assay medium (growth medium containing 2% FBS)", not the 10% growth medium — see §15. The cell count at exposure is still not stated for differentiated HepaRG; 2.2 × 10^5 is used. Because the measured fraction and the predicted fraction both scale with cell number, the ratio is insensitive to it.

9. Remaining caveats

- Fragki's 6.25 µM values in the project sheet (PFOA 0.0562, PFNA 0.281) still disagree with supplementary Table 2 (0.09226, 0.22203); the 25 and 100 µM values match to 1–3%. On the supplementary values Fragki PFOA gives b = 1.23 rather than 1.42, so the direction is robust.

- Armitage 2021 did run PFOS through IV-MBM EQP v2.0 (workbook sheets AEID767, AEID1325), which would make a direct validation target, but the sheets' two- row headers defeat clean extraction and the PFAS rows could not be read reliably. Not pursued.
- ~~Rosenmai's exposure FBS is unverified at source.~~ **Resolved 2026-10-20**: the paper is now in hand and §2.2 gives DMEM + 10% FBS with no serum reduction at exposure, so the coded value is right. What it also gives is a **3% ethanol** co-solvent (§2.1, §2.4) — see §15.
- AREc2's water content of 0.944 is high for a cell and its protein of 0.0514 low; it is used because it is the composition Armitage's own model was run with for a HepG2-derived line, not because it is independently plausible.

10. The fraction-of-dose axis, and the project spreadsheet on it

Figure 10 is now built on fraction of dose in cells (§7). Converting the spreadsheet's nmol/mg columns with the agreed 675 pg protein per cell:

over 29 points	bias	\	error\
project sheet Armitage	**0.90×**	2.86×	
published-equation Armitage	3.87×	**4.02×**	
published-equation Qin	7.52×	7.52×	
project sheet Qin	**57×**	57×	

The sheet's Armitage is essentially unbiased and runs 0.39× of the published-equation version on average — the two bracket the data.

The sheet's Qin is a different matter. Its bias and its absolute error are equal to four figures, so every point is high by nearly the same factor: a scale error, not scatter. On the fraction axis the failure is obvious — it places **6–82% of the dose inside cells that occupy 0.18% of the well volume**, where Rosenmai and Amstutz both measure 0.04–0.33%. Two candidates, in order of size:

1. **D_cell/w used as a lumped ratio** rather than composed as `D_SP/w × V_protein,cell + D_lip/w × V_lipid,cell`. For PFBA the lumped route gives 31 per litre of whole cell against the composed route's 2.4 — 13×.
2. **Dividing by C_nominal rather than C_free.** Qin's ratios are defined against the freely dissolved concentration. Small for PFBA (~1.25×) but large by PFOA, where albumin holds 89% of the dose.

Neither is visible in the sheet's output columns, so these are checks to run rather than a decomposition.

Rosenmai on the fraction axis

This is the cleanest form of the unit problem, because the axis is unit-free and the comparison is internal to one paper. At 100 µM:

	Table 1 uptake	Fig 2A as ng/mg	Fig 2A as nmol/mg
PFBA	0.100%	0.072%	15.4%
PFHxA	0.140%	0.064%	20.1%
PFHpA	0.210%	0.100%	36.3%
PFOA	0.330%	0.155%	64.2%
PFNA	0.310%	0.121%	56.0%

Across all 15 rows the nmol/mg reading contradicts Rosenmai's own uptake table by **141–215×**, and would put 15–64% of the dose inside 0.18% of the well volume. The ng/mg reading sits within 1.4–2.1× of the uptake table, the residual being the unstated working volume (§6b implies ~0.7 mL/well against the 1 mL assumed).

11. The wash artefact hypothesis: refuted on three counts

Earlier sections floated five ice-cold PBS/DPBS washes as a chain-length- dependent loss that would bias measured C_cell low. It does not survive.

(a) It is not a between-study variable. All three studies wash five times: Rosenmai "washed 5 times with ice-cold phosphate-buffered saline", Amstutz "washed... with DPBS" following Rosenmai's protocol, and Fragki "Cells were washed 5 times with Dulbecco's Phosphate Buffered Saline". A shared protocol cannot explain differences between them.

(b) The direction is backwards. Washing should strip the shortest, least bound, fastest-desorbing chains hardest, so the models should over-predict *short* chains most. They do the opposite — Armitage's over-prediction rises monotonically with chain length:

	nCF	Armitage/measured	Qin/measured
PFBA	3	1.42×	2.36×
PFHxA	5	1.20×	8.78×
PFHpA	6	1.55×	9.03×
PFOA	7	6.21×	12.71×
PFNA	8	8.54×	5.62×

Armitage: **+0.172 decades per CF2, r = +0.83**. Qin now shows the same trend (+0.092, r = +0.63); before the §15 correction it showed none. Washing would strip short chains hardest, so both trends point away from the hypothesis.

(c) The magnitude is two orders out. Fragki's time course puts steady state at ~1 h, so the cell↔medium exchange half-time is ~10–30 min. Five quick rinses give 1–2 min of contact:

exchange t½	1 min contact	2 min	5 min
10 min	6.7% lost	12.9%	29.3%
20 min	3.4%	6.7%	15.9%
30 min	2.3%	4.5%	10.9%

A 3–5× loss needs 1.5–2.3 exchange half-lives of contact — 15–45 minutes of washing. The hypothesis is dead at the order-of-magnitude level.

The one thing it was invoked to explain, Amstutz reading 3.6–4.8× below Rosenmai for PFHxA and PFHpA while agreeing to 1.0–1.4× for PFOA and PFNA, is better explained by short-chain analytical recovery: Rosenmai's own Table S2 reports internal-standard recoveries of 78–98% that vary by compound.

12. Testing the albumin parameter against the best measurement

Added 2026-10-16. Reproduce with `python3 tk_review/scripts/mbm_fischer_albumin.py`; full detail in `report/REPORT.md` §6.1b–6.1c.

Section 4 of this memo found the divergence between the two models to be in albumin, not in the cell. That raised an obvious question: can either model be improved by giving it a *measured* albumin partition coefficient instead of its own? Fischer 2024 is the only albumin dataset taken at a ligand:protein ratio below 0.004, so it is the natural reference. The assays run on fetal bovine serum, so its BSA column applies.

Converting a binding constant into a partition coefficient is exact in the trace limit — `D = 1000·n·Ka / M_alb`, so `log D = log Ka + log n - 1.823` (mass basis) or `- 1.688` (the volume basis both models use). Qin's own albumin term is this expression: identifying `Cmax = n × 20,527 µM` reproduces Qin's two Cmax columns to 0.3% on all four of their measured compounds.

Result: the substitution makes both models worse.

variant	bias	\	err\		<=25 µM	100 µM	dose swing
Qin, published	7.52×	7.52×	5.99×	8.63×	1.44×		
Qin + Fischer, constant D	7.75×	7.75×	10.39×	6.47×	**0.62×**		

variant	bias	\	err\		<=25 µM	100 µM	dose swing
Qin + Fischer, rescaled Langmuir	15.71×	15.71×	17.59×	14.66×	0.83×		
Armitage, published	**3.87×**	**4.02×**	6.08×	2.94×	0.48×		
Armitage + Fischer	8.08×	8.08×	12.01×	6.34×	0.53×		

Fischer's value is below both models' for every compound (0.14–0.61× of Qin, 0.17–0.61× of Armitage), so substituting it shrinks the albumin sink and pushes more dose into the cell — and both models already over-predict.

Two findings survive the negative result.

Qin's dose dependence is an albumin-saturation artefact. Its published error swings 3.49× across the dose range; with a constant albumin coefficient the swing is 0.99×. The dose exponent Qin produces (1.15–1.82 against a measured 0.85–1.42, section 4 of this memo) therefore says nothing about PFAS uptake — it is the Langmuir term coming off saturation.

Albumin cannot carry the level error. Inverting the balance for the D_alb that would make each model unbiased gives values **above** each model's current one, while the measurement sits **below** it. For PFHxA the needed value exceeds anything ever measured for that compound by 8.6–13.2×, even allowing six binding sites. So albumin is where the sensitivity is (it holds 19–94% of sorptive capacity) but not where the error is, and the remaining candidates are the cell term, the plastic term, and the measured C_cell values.

13. The cell term: slope exonerated, level convicted

Added 2026-10-19. Reproduce with `python3 tk_review/scripts/mbm_cell_term.py`; full detail in `report/REPORT.md` §6.1d.

Section 12 ruled albumin out as the source of the over-prediction. The cell term is the other large compartment, and it is where the two models disagree most about mechanism while agreeing most about value.

Composition. Armitage's cell is 79% water at PFBA and 99.7% membrane lipid at PFNA, with protein contributing 0.1–0.2% throughout (only 3.5% of protein is sorptive, and that fraction is treated as octanol). Qin's cell is 92–95% protein at every chain length, with no water phase. They differ on the cell partition by 0.44–9.5×, yet their predicted fraction of dose in the cell agrees to 0.6–2.5×, because the albumin term damps the difference. Agreement on the output is not corroboration of the mechanism.

The slope. Against Qin's own Table 2 both models look 2.9–4.4× too steep in chain length. That reading is wrong. Inverting the balance for the cell partition the uptake data require gives **+0.407** decades per CF_2 on the corrected media; the models give +0.515 and +0.782; Ryu 2024's whole-tissue f_unbound gives 0.46–0.53. Qin's Table 2 alone gives +0.177, and at PFBA it is 20× above what the uptake requires. Three independent lines against one: Qin's slope is fine. **Armitage's is not** — it sits 0.38 above the uptake requirement, and §16 shows that is where the residual chain-length dependence comes from.

Substituting Qin's measured D_cell/w confirms it. Mean bias improves (Qin 7.52 -> 4.61, Armitage 3.87 -> 2.15) and so does absolute error (7.52 -> 5.43, 4.02 -> 2.91) — but look at where it goes: PFBA is over-predicted **19.7× and 23.6×** while PFNA lands at 1.14× and 0.73×. The aggregate improves because a wild over-prediction at one end is being averaged against a good fit at the other. Centring the fit while destroying the trend is what substituting an outlier looks like.

The level. Qin's cell partition runs 1.5–9.2× above the uptake requirement and Armitage's 0.25–6.2×, both worst mid-series at PFHxA–PFHpA. Those factors are conditional: redoing the inversion under Qin's, Armitage's and Fischer's albumin moves the implied level 2.6–8.3×, though the slope survives all three (+0.72 to +0.82). Read the level as "several-fold too high, worst mid-chain".

What to fix. Not the lipid or octanol descriptor — the chain-length dependence is already right. The two remaining candidates are the sorptive capacity assigned to cell protein (Qin's measured D_non-specific versus Armitage's "3.5% of protein behaves as octanol" — wildly different guesses at the same quantity, and the substance of each model's cell term) and the cell volume the result is normalised by, which is uncertain to 3.1× across the three compositions in play.

14. Candidate 1 eliminated; the cell volume is what is left

Added 2026-10-19. Reproduce with `python3 tk_review/scripts/mbm_cell_protein.py`; detail in `report/REPORT.md` §6.1e. This closes the mass-balance line.

Two papers made the test possible. **Qin et al. 2024** (ES&T 58:9954, the same group as the 2023 model) measured specific and nonspecific protein–water distribution ratios for 16 PFAS by SPME over four orders of magnitude. **Chen et al. 2025** (ES&T 59:82) independently writes the Kd -> partition coefficient bridge of §6.1b as its Equation 4.

Candidate 1 was cell protein. Qin's 2023 interpolation turns out to sit within 1.1–2.5× of his own later measurement with the same slope, so the model was not damaged by interpolating three of the five compounds. Armitage's octanol surrogate is another matter: 0.035·D_ow runs **457–7,820× below** the measured protein affinity and 1.5× too steep. That is why Armitage's cell is ~100% membrane lipid — the convention has defined its protein term out of existence.

But substituting the measured value makes both models **worse** (Qin 7.52 -> 12.24, Armitage 3.87 -> 9.81), because the measurement is *higher* than either model assumes and both already over-predict.

The pattern across all three substitutions:

parameter	best measurement	direction	effect
albumin (Fischer 2024)	below both models	smaller competing sink	worse
cell D_cell/w (Qin 2023 T2)	mixed	—	centres, disperses
cell protein (Qin 2024)	above both models	bigger cell sink	worse

All three move dose into the cell. No partition coefficient fixes an over-prediction here, because the measured values do not point that way.

The cell volume halves it, and no longer closes it. Rerunning the published models at 3.04 pL/cell (project sheet, HepG2) instead of 10 ng/cell (AREc2) gives Qin 7.52 -> 3.71× and Armitage 3.87 -> 1.84×. It is still the one knob that rescales every compound by the same factor, which is what a level offset independent of chain length requires — and the assays modelled are HepG2 and HepaRG, not AREc2. But before §15 this substitution reached 1.60× and 1.08×, i.e. it looked like the whole answer. On the corrected medium it accounts for about half the residual in log terms and leaves a 1.8–3.7× over-prediction standing. None of the three uptake papers reports its cell volume, so that part stops here; what §15 adds is that the cell volume now has to share the blame.

One result worth keeping regardless. Qin 2024 reports both binding regimes for the same protein and compound: specific exceeds nonspecific by 3.0–5.0× across PFBA–PFOA, with specific dominating below *nu* ~ 0.01 and nonspecific above *nu* ~ 1. That is the direct measurement of the ligand:protein regime effect §6.1a inferred, and it quantifies the penalty for carrying an in vitro constant into a physiological calculation without asking which regime produced it.

15. The exposure medium is not the growth medium

Added 2026-10-20. Reproduce with `python3 tk_review/scripts/mbm_medium_composition.py`; output `db/mbm_medium_composition.csv`.

All three assays were coded at 10% FBS, read off each paper's **culture** section. Two of the three state something different for the medium the cells were actually **dosed** in.

	growth medium	exposure medium	DMSO carrier	verified
Rosenmai 2018	DMEM + 10% FBS	*the culture medium*, 10% FBS	**3% ethanol**	yes
Amstutz 2022	DMEM + 10% FBS	*the culture medium*, 10% FBS	0.1% DMSO	yes
Fragki 2022	William's E + 10% FBS	**assay medium: growth medium + 2% FBS**	0.5% DMSO	yes

Fragki et al. 2022, §2.2–2.3, verbatim:

> *"Prior to biokinetic studies, differentiated HepaRG cells were incubated for 24 h in assay medium (growth medium containing **2% FBS**) supplemented with 0.5% DMSO… Test chemicals were diluted from 200-fold concentrated stock solutions **in assay medium**, providing exposure medium with a final DMSO concentration of 0.5%."*

Amstutz et al. 2022, §2.3, verbatim: PFAS were "added from stocks in DMSO **to the culture medium**", the DMEM + 10% FBS of §2.2.

Why it matters more than it looks. Serum fraction multiplies the two largest sinks in the denominator:

```
V_albumin     = V_medium · f_FBS · 0.733 mL/g · 42 g/L
V_serum_lipid = V_medium · f_FBS · 1.03 g/L
```

and in Fragki's well albumin carries **97% of the denominator for PFOA** at 10% FBS. Cutting the serum five-fold shrinks the sorptive capacity ~4×, which raises the free concentration ~4×, which raises every predicted uptake ~4×.

Fragki, bias	at 10% FBS (as coded)	at 2% FBS (as dosed)
Qin	1.54×	**14.92×**
Armitage	2.48×	**10.80×**

The dataset that looked like the models' best case was the one being handed five times more albumin than it had. Overall: Qin 3.43 -> 7.52×, Armitage 2.33 -> 3.87×.

Armitage moves by 4.4×, which is the denominator argument almost exactly, because its albumin term is linear. Qin moves by 9.7× — *more* than the denominator shrinks — because its albumin is a saturating Langmuir: a higher free concentration drives D_BSA/w down, which shrinks the denominator again. **A model with a saturating sink amplifies an error in the size of that sink.** That is a general property worth carrying forward, not a quirk of this dataset.

The co-solvents: real, 30-fold apart, and almost worthless

The three assays dose from three different vehicles, and the spread is larger than the FBS spread looks:

	co-solvent	*sigma*	log shift	as a factor
Rosenmai	**3% ethanol**	4.5	**0.135**	**1.36×**
Fragki	0.5% DMSO	4.0	0.020	1.05×
Amstutz	0.1% DMSO	4.0	0.004	1.01×
the serum difference	—	—	*0.70*	*5.00*×

Log-linear cosolvency shifts log K by *sigma*·f_cosolvent; *sigma* ~ 1.5–4 per unit volume fraction for a hydrophobic solute in aqueous DMSO and a little more in ethanol, and the aggressive end is used throughout so the estimate cannot flatter the conclusion. Rosenmai's ethanol is **7× Fragki's DMSO shift and 34× Amstutz's**. So the co-solvent *is* a between-study variable — just not the one the 2% vs 10% looked like, and still 5× smaller than the serum.

And it cancels. A co-solvent lowers *every* water-to-phase partition coefficient by the same factor k, and the quantity being predicted is a ratio of sorptive terms:

```
x_cell = D_cell·V_cell / (V_w + D_alb·V_alb + D_lip·V_lip + D_cell·V_cell)
```

Divide every D by k and k cancels top and bottom — except against **V_w**, the one term with no D on it. So the co-solvent only bites where water carries a real share of the denominator. Dividing all of Rosenmai's D by 1.35:

	Armitage bias	PFBA	PFHxA	PFOA	PFNA
as published	1.97×	1.42×	0.86×	2.38×	9.33×
÷1.35 (3% ethanol)	1.86×	**1.11×**	0.83×	2.37×	**9.31×**

PFBA moves (water is 80% of its denominator); PFNA does not move at all (water is 0.5% of its denominator). **The general lesson is worth more than the number: an error that multiplies every sorptive term cancels on this axis, and only an error in the *relative* size of two sinks survives.** That is exactly why the FBS fraction is devastating and the vehicle is not.

One DMSO point is *not* about partitioning. HepaRG are differentiated in 1.7% DMSO for twelve days and held there — that is what makes them hepatocyte-like, and it also loads them with triglyceride, which the companion paper (Louisse et al. 2023) measures as an endpoint. Both models run Fragki on the **AREc2** composition, a HepG2-derived reporter line with membrane lipid 0.00414, because that is the only liver-line composition Armitage's SI publishes. A differentiated, DMSO-matured HepaRG is a different cell. That error is in the cell term, which §13–14 already name as where the residual lives — so the two errors are independent and stack.

Does the measured data show the serum effect?

If 2% FBS leaves five times less albumin competing with the cells, Fragki's *measured* uptake should sit above the 10% FBS studies at the same nominal dose. It does:

	dose	Rosenmai 10%	Amstutz 10%	Fragki 2%	ratio
PFOA	100 µM	0.330%	0.245%	0.780%	2.7×
PFNA	100 µM	0.310%	0.302%	4.385%	14.3×

The direction is right and the PFOA magnitude is close to the predicted ~4×. It is corroboration, not proof: the cell line differs too, and differentiated HepaRG are larger and more lipid-rich than HepG2. PFNA at 100 µM is also the one point Fragki themselves set aside from their QIVIVE, for low protein content suggesting cytotoxicity; it is retained in the model comparison here and it is the largest measured value in the set.

What this changes

1. **§5 performance.** Both models over-predict everywhere. PFBA–PFHpA, which only the 10% FBS studies measure, are untouched (Armitage 1.2–1.6×). PFOA and PFNA carry almost the whole residual.
2. **§4 dose exponent** *improves* for Qin on Fragki, 1.71 -> 1.49 against a measured 1.42. Some of the curvature Qin was adding was an artefact of a sink five times too big.
3. **§14's conclusion weakens.** The cell-volume substitution used to reach 1.08× for Armitage and looked like the whole answer; on the corrected medium it reaches 1.84× and accounts for about half the residual.
4. **A methodological point for the QIVIVE literature.** Fragki's biokinetics were explicitly "designed based on the study of [Amstutz], who assessed the cellular accumulation of PFASs in HepG2 cells", and the two datasets get used side by side. They were measured five-fold apart in the single medium property a mass balance is most sensitive to, and neither paper flags it.

Still open. All three media are now read at source. What is not in hand: Fragki and Amstutz both took a t = 0 medium sample for a mass balance (550 -> 500 µL and 250 -> 200 µL) and neither mass balance is reported in the text available here. If it were, it would test the plastic term directly instead of by difference. Rosenmai reports 78–98% recovery in the cell lysates but no medium-side balance either.

Also unverified rather than wrong: Rosenmai's §2.5 states the 12-well format, 18×10^4 cells per well and three pooled wells — all as coded — but **not the working volume**, which remains the 1 mL/well assumption of §8. And the 3% ethanol is stated in §2.4 for the PPAR*alpha* assay; §2.5 does not repeat it, though it doses the same ethanol stocks at the same concentrations.

!three datasets

16. The residual rises with chain length, and that localises it

Added 2026-10-20. Reproduce with `python3 tk_review/scripts/mbm_chain_length_residual.py`; output `db/mbm_chain_length_residual.csv`.

After §15, Armitage over-predicts by 3.87× and Qin by 7.52×. A pure level error would sit flat across the series. It does not: Armitage's climbs **+0.172 decades per CF_2, r = +0.83**. Four questions, each able to kill a candidate outright.

Q1. Is the trend real, or an artefact of pooling three studies?

Fragki measures only PFOA and PFNA and carries the biggest residuals, so pooling alone could manufacture a slope. Ask the question inside one lab:

	PFBA	PFHxA	PFHpA	PFOA	PFNA	slope	r
Rosenmai — Armitage	1.42×	0.86×	1.10×	2.38×	9.33×	**+0.150**	**+0.70**
Rosenmai — Qin	2.36×	5.27×	5.09×	3.23×	3.48×	+0.022	+0.29
Amstutz — Armitage	—	3.33×	4.30×	2.38×	5.93×	+0.050	+0.38
Amstutz — Qin	—	40.5×	50.5×	9.47×	6.65×	**-0.308**	-0.90

The trend is Armitage's, and it is real. Rosenmai holds one lab, one cell line, one medium, one vehicle and five compounds fixed, and it is still there. Qin has no trend inside a study — positive at Rosenmai, *negative* at Amstutz — so Qin's pooled +0.092 is precisely the pooling artefact this test was built to catch.

So the question sharpens: **why does Armitage's over-prediction grow with chain length where Qin's does not?**

Q2. Is it cytotoxicity at the long-chain end?

Long chains are the cytotoxic ones. Rosenmai report a significant drop in protein content at 100 µM PFDA and PFTeDA; Fragki excluded PFNA at 100 µM from their own QIVIVE for the same reason. Dying cells losing PFAS — or losing the protein the uptake is normalised to — would read as over-prediction at the long end. That hypothesis makes a sharp prediction: worst at the highest dose.

Rosenmai, Armitage	PFBA	PFHxA	PFHpA	PFOA	PFNA	slope	r
10 µM	1.24×	0.77×	1.34×	2.75×	13.50×	**+0.194**	+0.77
30 µM	1.34×	0.77×	0.97×	2.44×	9.20×	+0.157	+0.70
100 µM	1.74×	1.05×	1.02×	2.00×	6.53×	**+0.099**	+0.58

Refuted, and backwards. The trend is *strongest* at 10 µM, where nothing is dying, and weakest at 100 µM. It joins the wash hypothesis (§11) on the list of measurement artefacts that fail their own direction test.

Q3. Is the fitted slope itself an artefact of PFBA?

PFBA sits at nCF 3, two carbons from the next compound in a five-point regression, so it swings the fitted line hard. It is also the one compound where **water** carries the denominator — 80% of it, against 0.5% at PFNA — which makes it the point least informative about the sorptive terms. Reporting every slope twice:

Armitage, Rosenmai	PFBA	PFHxA	PFHpA	PFOA	PFNA	all five	nCF 5–8
	1.42×	0.86×	1.10×	2.38×	9.33×	+0.150	**+0.345**

The trend is more than twice what the five-point fit says. PFBA sits *above* the line at the short end and drags the slope down. Over nCF 5–8, Armitage climbs 0.86× -> 9.33×, a factor of **10.8 across three CF_2**. Qin over the same range: -0.074, i.e. nothing. The contrast sharpens once the leverage point is out.

Q4. Which term carries it?

§12–14 judged three substitutions on the **level**, where all three made the fit worse. On the slope (Rosenmai):

Armitage variant	all five	nCF 5–8
published	+0.150	**+0.345**
+ Fischer 2024 albumin	+0.212	**+0.239**
+ Qin 2024 cell protein	+0.067	**+0.248**

Armitage variant	all five	nCF 5–8
+ Qin 2023 measured D_cell/w	-0.321	**-0.243**

No single substitution zeroes it. Albumin takes out about a third, cell protein about a third, and the cell partition over-corrects past zero.

> *Correction to an earlier draft of this section: the five-compound column says Fischer's albumin makes the slope **steeper**, and I read that as evidence the albumin term was innocent. It is the PFBA trap. Fischer's albumin is 10× below Armitage's at PFHxA but only 2.9× below at PFNA, so the short chains gain most — except PFBA, which is water-dominated and barely moves, tipping the five-point line the other way. Over nCF 5–8 the albumin substitution **flattens** the trend, as it should.*

Q5. The mechanism: it is the *ratio*, not either term

Once albumin dominates the denominator — 94–97% of it from PFOA up —

```
x_cell  ~  D_cell · V_cell / (D_alb · V_alb)
```

so the residual's chain-length slope is the chain-length slope of the **ratio D_cell/D_alb**. Neither coefficient alone predicts it.

	nCF	Arm D_alb	Arm D_cell	ratio	Qin D_alb	Qin D_cell	ratio	Fischer BSA
PFHxA	5	3.81	1.00	-2.81	3.83	1.40	-2.43	2.83
PFHpA	6	4.42	1.75	-2.67	4.38	1.92	-2.46	3.93
PFOA	7	4.71	2.53	-2.19	4.93	2.44	-2.49	4.20
PFNA	8	5.00	3.31	**-1.69**	5.48	2.95	**-2.52**	4.54

Decades per CF_2 over nCF 5–8:

	albumin	cell	ratio	residual, observed
Armitage	+0.386	+0.769	**+0.383**	**+0.345**
Qin	+0.547	+0.515	**-0.032**	**-0.074**
Fischer 2024, measured BSA	**+0.540**	—	—	—

The ratio *is* the residual, to within the noise on a four-point fit.

Armitage gets it wrong from both ends, and the two errors stack:

- its **albumin rises at +0.386** where Fischer measures **+0.540** — too shallow by 0.154, so the albumin sink grows too slowly with chain length;
- its **cell rises at +0.769** where a flat ratio needs ~ +0.540 — too steep by 0.229.

40% of the gap is the albumin spLFER, 60% the cell term. Both have the same root: the piecewise spLFER flattens to 0.37·log K_ow above log K_ow 4.5, and the cell is 99.7% membrane lipid at PFNA because *"3.5% of cell protein behaves as octanol"* puts the protein term 457–7,820× below the measured affinity (§14). **A cell made of lipid climbs faster with chain length than a cell made of protein; an albumin spLFER that flattens climbs slower than albumin does.**

Qin gets both right. Albumin +0.547 — within 0.007 of Fischer's measured +0.540 — and cell +0.515, so the ratio is flat to 0.032 per CF_2.

> ***Qin has the right chain-length physics and the wrong level; Armitage has the better level and the wrong chain-length physics.*** *The two failures are independent, which is why no single substitution fixes both, and why their aggregate biases keep swapping places with dose and with compound set.*

What would settle it

A measured D_cell/w for PFBA->PFNA on **HepG2 or HepaRG** rather than HEK293H homogenate, at a trace ligand:protein ratio, read alongside a measured albumin series on the same compounds. **The quantity to report is the ratio and its slope**, not either coefficient alone. The existing ledger for the cell term — Qin's Table 2 at +0.177, the uptake inversion at +0.407, Ryu 2024's whole-tissue f_unbound at +0.46 to +0.53, and the two models at +0.515 and +0.782 — brackets it between roughly +0.4 and +0.55 without pinning it.